

A Machine Learning based Approach to Predict the Effects of Video Games on Youth Behavior

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Abstract— Video game addiction is a rapidly growing concern among young people, emerging as a major issue due to its harmful impact on their mental health. This study investigates how predictive models might reveal subtle patterns in gaming habits by examining the relationship between video games, youth development, and machine learning. The research will investigate social dynamics as well as cognitive, academic, and mental health implications in order to educate individuals about responsible gaming habits and teaching tactics. The study covers ethical considerations and long-term societal ramifications, emphasizing the significance of a compatible way to navigating the IT based world because of young people's prosperity. Its purpose is to evaluate whether a gamer is career-minded. That's why we gathered 804 data points from them, of which 780 were utilized. The key point that we utilized were name, gender, age, amount of time spent studying, amount of sleep or wakefulness, amount of time spent playing games, reason for playing games, amount of time spent playing games more than with family, time spent honing skills, and time spent worried about a career. and after that, they underwent reprocessing and were checked before being used with certain machine learning algorithms. Various prediction and find ways use machine learning, artificial intelligence, and deep learning method. Our approaches include Gaussian Naive Bayes (GNB), Random Forest (RF), Adaptive Boosting (ADA Boosting), Decision Tree (DT), Extra Trees (ET), Gradient Boosting (GB) classifier, KNN (K-Nearest Neighbors), Support Vector Machine (SVM). The Random Forest classifier performed better in our analysis than the other eight methods. While all eight classifiers have great prediction accuracy, Random Forest Classifier (RF) has the highest accuracy of 97.00%. This study analyses the various behaviours or habits of the current youth society. Based on the results obtained from that analysis, a machine learning-based model has been proposed. This proposed model will help in understanding the behavior of young people and their current situation.

Keywords— *Machine Learning; Video Games Addiction; Impact on Video Games; Online Gaming; Random Forest.*

I. INTRODUCTION

The internet has become a vital part of everyday life and has grown exponentially [1]. Gaming in online has grown a famous

source of entertainment, drawing individuals of all ages and backgrounds. However, this broad trend has a negative impact on human connections and relationships. Excessive internet gaming can damage relationships and lead to neglect of vital obligations. In certain situations, people can get so engrossed by gaming that they entirely disregard their responsibilities and recently recognized as mental health disorder [2]. Thus, while internet gaming provides universal delight, its excessive use can considerably undermine personal relationships and impede the fulfilment of tasks in contemporary society [3].

Video games have experienced unprecedented growth in popularity and accessibility in recent years, changing the way people consume entertainment, particularly young people. The debate over the gaming industry's potential consequences on several aspects of society, such as adolescent people's cultural, psychological, and mental growth, is always evolving [4] [5].

During these discussions, one notable topic of research emerged: the use of ML to forecast the effect of gaming on young people. This research attempts at the connection between video games and machine learning [6], with a focus on forecasting the effects on young people. Researchers want to use machine learning algorithms to evaluate complicated trends associated with gaming behaviors, habits shifts, and possible linkages between social connections, mental health, and academic success since ML techniques [7] and data mining procedures have been effectively used for analysis of multidimensional datasets [8]. The prediction model is developed utilizing machine learning approaches provide insights into how video games may affect young people. Forecast for Video games have an impact on the younger generation by providing a mechanism for determining a person's mental health [9], such as whether or not they are anxious about their work. We are using certain data like person's Name, age, amount of time spent studying, resting, waking up, amount of time spent playing games, the reason for playing game, how much time is spent honing skills. We tried to find the proportion of boys and girls addicts after analyzing our information and also offer a solid structure for forecasting results and find a trends the impact of gaming on society and gamer.

II. LITERATURE REVIEW

Video games has grown an integral section of daily life for millions of people, particularly the younger generation, in an era driven by technical breakthroughs and digital immersion. With an emphasis on the younger population, the increasing popularity of video game culture raises important concerns about how it affects people's cognitive, social, and emotional development. As long as these issues are unanswered, machine learning will be a useful tool in analyzing the complex connections between young people's gaming behaviors and their effects. This section discusses recent and significant literature on video games addiction. We observed and examined their work to more knowledge their procedures and strategies. This part, we'll talk about previous research related to our topic.

In this document [10], they attempted to predict whether a certain person was addicted to video games. Since consequence, Google Forms was used to collect 549 data. During the data collection process, a seven-criteria gaming addiction measurement scale was used. After testing numerous algorithms for machine learning, it was determined that adaptive boosting, or ada boosting, outperformed other approaches that have a 93% accuracy. Computations found that 25.0% of persons suffer from a game playing.

This research [11] discusses how online gaming can have a negative impact on learning outcomes and concentration. Classification methods were employed to find. Naive Bayes algorithm is more accurate than RF and C4.5. The NB method can predict student academic performance. NB algorithm is more accurate than RF and C4.5. The NB method achieves an accuracy that is 69.18% and an AUC grade that is 0.77. The RF algorithm is 66.34% accurate, with an AUC value of 0.74. The C4.5 achieved an accuracy that is 65.65% and an AUC value of 0.686.

In this research [12] the work looks at negative impacts of extra gaming with video, with a focus on Asian PUBG players. The likelihood of human problems in Asian PUBG players was calculated using supervised ML. The classification method included plotting the ROC curve, computing AUC, growing iterations, identify variables, and giving accuracy in %. The Logistic Regression classifier predicted IGD and ADHD with a highest accuracy those are 93.18% and 81.81%, gradually. The Decision Tree algorithm was the highest accurate finder of GAD at 84.9%. The work acknowledges like key limitation the potential of biases, like social attractiveness bias, in report data evaluation, that might lead to an over or under-reporting of psychiatric issues. The document [13] analyses negative effects of too much gaming and find danger variables for the risk of unhealthy gaming habits. It uses a long term research with 565 gamers and uses AI classifiers to predict gaming disorder (GD) risk. Random Forests outperformed all other AI's, with avatar immersion being the strongest predictor where [14] showed GD is linked with decline in self-esteem and also deficiencies in the area of physical, social and emotional self-concept.

The paper uses functionalist perspective to propose a balanced model for antecedents of online game addiction among adolescents. It identifies functional needs like relationship and escapism driving addiction, like education and

attention switching activities. The study suggests that motivation and prevention factors are partially mediated by online gaming [15] [16].

The author [17] used a LCA to classify different set of survey respondents. According to this report, 1.5% of 13-16-year-olds and 3% of gamers are addicted to the internet. The major goal of this study was to explore association among mental health and video game addiction. While authors [18] used Facial Expression Recognitions (FER) and Emotion Recognition (SER) method to figure out the change of behavior during game.

This work aims to assess the outcome of internet gaming on student's academic progress and social behavior. This study aimed to examine the result of online games on student's cognitive abilities, academic performance, and social behavior. We use machine learning techniques to determine the optimal score and identify the most addictive age group based on survey data.

III. METHODOLOGY

Machine learning algorithms like random forest, Naive Bayes, K-nearest Neighbors [19], Gradient Boosting, Ada boost, Extra trees, SVM, and Decision Tree [20] can accurately predict the impact of video games on youth. The accurate result can be found consist of four phases such as (i) Pre-processing (ii) Model Selection (iii) Training and Testing data (iv) Classification. The data collection and pre-processing phase involves selecting data for training and testing our ML model, applying the model to determine the best algorithm for optimal results. The entire framework of this paper can be described with figure 1.

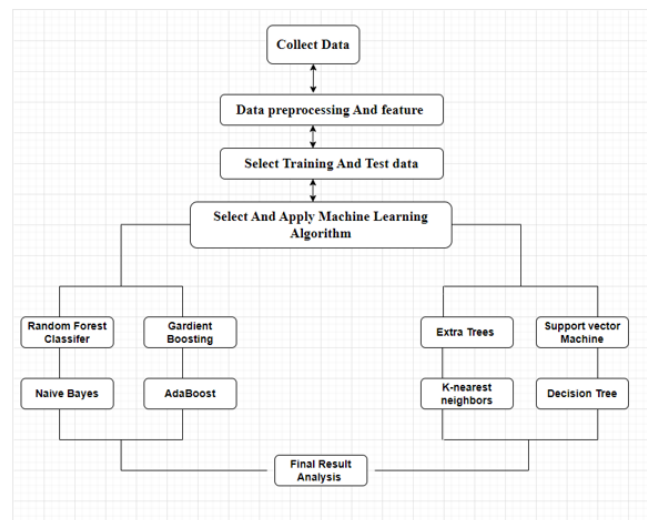


Fig. 1. A machine learning based proposed model

A. Dataset

In our experiment, the analysis is mainly offline and online based. We gathered data from public and private university students. Most data is collected through Google Forms, with some being collected physically. More than 800 data are accumulated containing 754(83.85%) numbers of male and 126(16.15%) number of female record but 780 of them are

used We have collected almost 800+ data but in our research we used 780 of them to enhance model accuracy. We splitted dataset into train set and test set. The dataset consists of 780 records, with 80% train and 20% test data, and uses various machine learning techniques to enhance model accuracy. We would also like to go into great depth about each component of our suggested approach so that everyone can understand it.

B. Data Pre-processing

For classifying the data, we set different attributes for datasets. The datasets contains 780 records with 654 men and 126 women. We have also attempted to determine the proportion of men and women who are anxious about their careers in our sample. The gender based summary is given in figure 2.

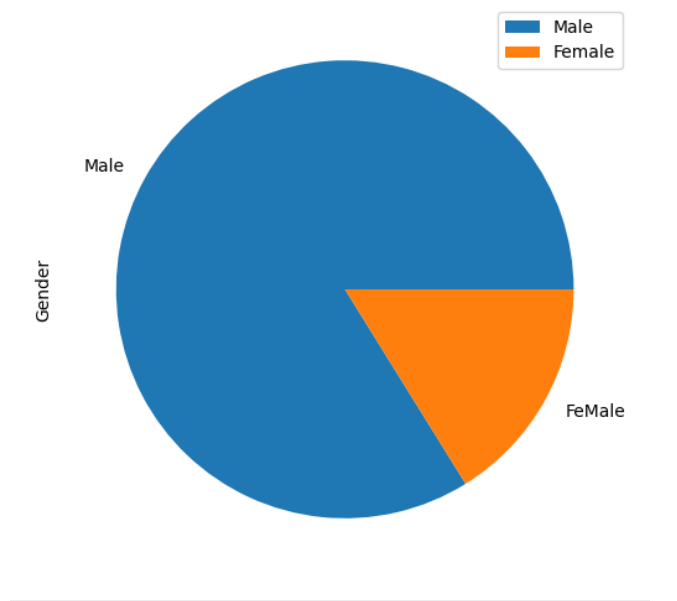


Fig. 2. Gender Based summary

The data, which focused on variables like name, gender, age, and amount of time spent playing games, wake-up time and sleeping time was branches into two sections for training sections and testing sections. The training and testing data using attributes Figure 3.

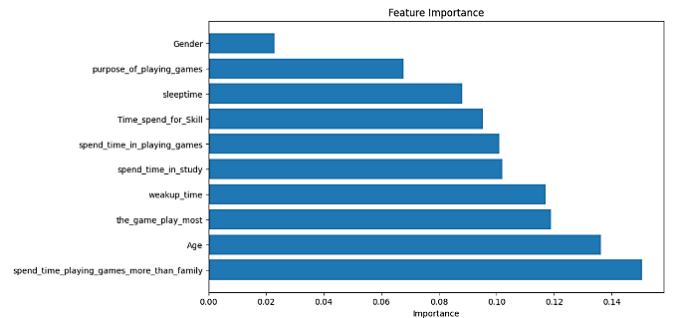


Fig. 3. Features of the dataset

A few values are missing. Therefore, we employed mean imputation, which functions as follows: whenever a missing

value is discovered, it is replaced with the mean value of all the situations in which it is present. After that we are able to handle the missing data. The figure 4 the missing data and the data after getting replaced.

C. Feature Engineering

As soon as the data was pre-processed, we were interested in finding out if every feature was equally important in predicting our target class. Therefore, in order to obtain this data, we used feature engineering techniques to identify the most crucial features. To identify the best features for class prediction, tree-based classifiers and Pearson correlation were employed.

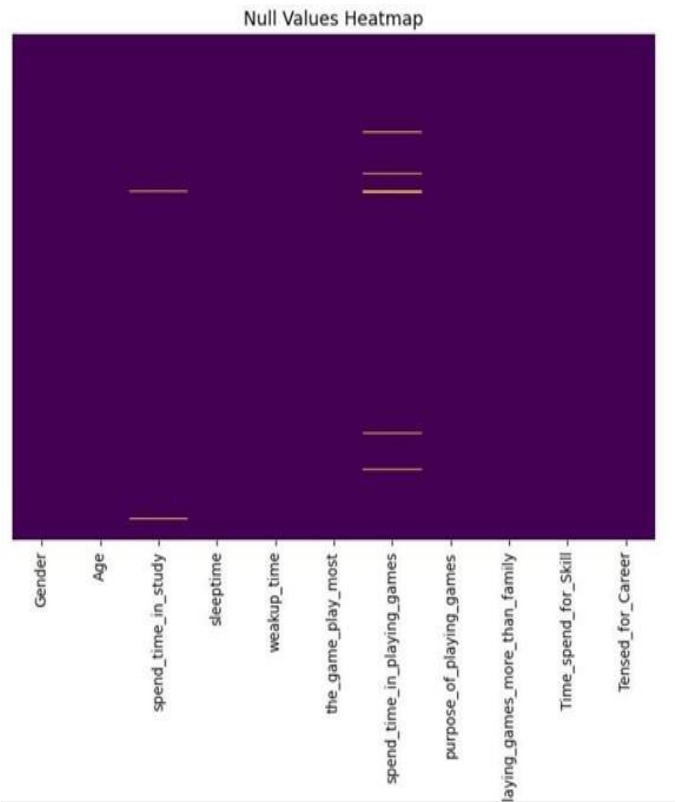


Fig. 4. Missing Data pre-processing and handling

D. Model Training

Several machine learning techniques that make use of machine language processing are available for estimating the influence of video games on youth analysis. We use nine types of machine learning classification algorithms i.e. Decision Tree (DT), Random Forest (RF) [21], Gaussian Naive Bayes (GNB), XGBOOST (XGB), Support Vector Machine (SVM) [22], Gradient Boosting (GB), Extra Trees, Adaptive Boosting (ADA BOOSTING) and K-Nearest Neighbor (KNN) to train our model [23].

E. Confusion Metrics

A table used to explain a classification model's performance is called a confusion matrix. It displays a summary of the number of true positive, true negative, false positive, and false negative predictions made by a model on a test dataset. It's a tool for figuring out how well a classifier is doing in terms of the many mistakes it makes.

True Positives (TP) are events with a positive correlation that the model correctly classifies as negative. True Negatives (TN) are events where the model classifies as negative even though they are in fact negative. False positives (FP) or Type I errors are the number of cases model specify the negative events as the positive one. False Negatives (FN) are positive cases that the model mistakenly classifies as negative (Type II error).

F. Evaluation Metrics

Evaluation metrics are the measurements tools which are used to test the efficacy and performance a statistical or machine learning model's. These tools help to make decision by comparing the performance of various models or algorithms. We have used three most popular evaluation metrics. These metrics estimate the precision of predicted classification with respect to actual classification.

These are precision, recall and f-measure. These metrics assist users in identifying best classification model of very high preference from the range of datasets available [24].

Precision: Precision is a measure of accuracy that assesses the percentage of relevant items from all items retrieved [25]. This measure should be as high as possible. It is computed as:

$$Precision = \frac{n}{topN} \quad (1)$$

We can also demonstrate it like:

$$Precision = \frac{true\ positive}{true\ positive + true\ negative}$$

In terms of classification, where a positive impact along with actual positive effect of the video , and positive impact along with actual negative effect game of vedio game are predicted by using the models., indicating that the game has no positive effect despite the prediction. Instances where the model predicts a negative impact from the video game, but the actual impact is positive, indicating a missed opportunity to identify a beneficial effect. Instances where the model predicts a negative impact from the video game, and the actual impact is also negative, suggesting a correct identification of non-beneficial effects.

Recall: To predict the effect of video games on youth, recall serves as a metric to gauge the model's ability to capture the proportion of positive effects identified among all the actual positive impacts of video games on young individuals. If M_T denotes the number of positive impacts recognized by users in the test set, and N stand for the number of positive impacts correctly identified and recommended by the model, recall elucidates the model's effectiveness in comprehensively recognizing the beneficial effects of video games among all such instances.

$$Recall = \frac{N}{M_T} \quad (2)$$

Another way of demonstrating recall is:

$$Recall = \frac{true\ positive}{true\ positive + false\ negative}$$

a) *F-measure*: The f-measure metric combines Precision and Recall into a single value for comparison purposes. It is used for a more balanced view of performance.

$$F_{Measures} = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (3)$$

IV. RESULT ANALYSIS AND DISCUSSION

In our data analysis, we categorize our dataset into two types of variables: X, representing all feature values, and Y, representing the target value. We utilize these variables to test our data and assess the system's performance using various evaluation indicator i.e. accuracy, precision, recall, F1-score. These metrics are used to evaluate different model's predictive capability and its effectiveness in capturing different aspects of the data's classification performance.

Eight classification techniques were employed to train the data, which divided into two sections: Training and Test, with 80% and 20% of the data in each section. The outcomes of models, used to classify the data are included in Table 1.

TABLE I. PERFORMANCE TABLE OF THE DIFFERENT CLASSIFIER

Algorithm Name	Accuracy	Precision	Recall	F1-Score
Random Forest Classifier	97.4	97.51	97.43	97.37
K-Nearest Classifier	82.69	81.18	82.69	79.81
Gradient Boosting	87.82	88.54	87.82	86.10
Gaussian Naive Bayes	76.92	71.54	76.92	72.77
Adapting Boosting	83.33	82.43	83.33	80.33
Extra trees	96.15	96.33	96.15	96.00
Support Vector Machine	78.84	83.32	78.84	69.52
Decision tree	94.23	94.13	94.23	94.12

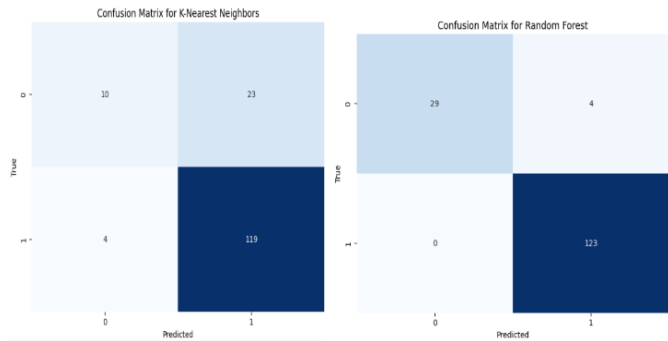


Fig. 5. Random Forest and K-Nearest Neighbors Confusion Matrix

According to the accuracy table, Random Forest Classifier (RF) comes up with accuracy of 97.4%, while K-Nearest Neighbor (KNN) with 82.69%, Gradient Boosting (GB) with 87.82%, Gaussian Naive Bayes (NB) is 76.92%, Adapting Boosting (ADA Boosting) with 83.33%, Extra Trees (ET) with 96.15%, Support Vector Machine (SVM) with 78.84%, Decision Tree (DT) with 94.23% respectively. Random Forest Classifier achieves the highest accuracy among all the algorithms.

The confusion matrices of each classifier are displayed in Figure 5-8 for a clearer understanding of the performances of our model.

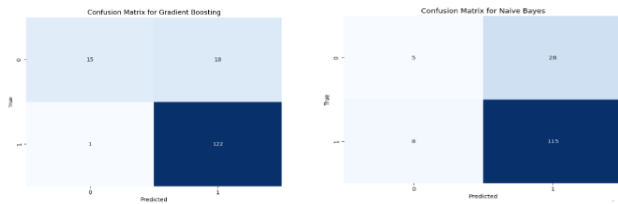


Fig. 6. Gradient Boosting and Naive Bayes Confusion Matrix

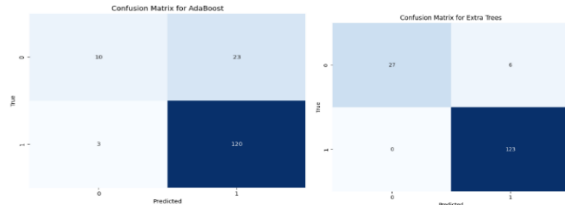


Fig. 7. Ada Boosting and Extra Trees Confusion Matrix

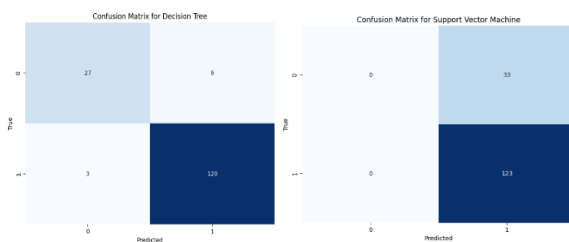


Fig. 8. Support Vector and Decision Tree Confusion Matrix

V. CONCLUSION

In today's technologically evolved society, technology is easily accessible to anyone, particularly the younger generation. Online gaming has become increasingly popular among today's youngsters due to simple access to the internet and cellphones. Today internet is widely accessible for the youth society due to the cheap and outnumber smartphones, as opposed to computers. Many people begin using smartphones as children become so engrossed in them that they lose touch with others.

Smartphone use often begins in childhood, leading to a lack of interaction with others. They grow addicted to the virtual world because it is more appealing than the real world. Addiction is most prevalent among individuals aged 20-25.

People are glued to social media platforms i.e. Facebook, Twitter, YouTube, and video games. Video games accounts for the majority of people's time. Video games have educational value, with many games designed to develop creativity, problem solving ability, critical thinking power and resource management efficiency of the youth. Educational games have proven to be effective tools in classrooms, engaging students and enhancing learning outcomes in subjects ranging from mathematics to history.

However, it's important to acknowledge the potential negative impacts of excessive gaming. Addiction to video games can lead to detrimental consequences of mental and physical health, and strained relationships with family and friends. Furthermore, there are concerns about the desensitization to violence that some games may perpetuate, as well as issues related to excessive screen time and sedentary

behavior. To understand the growing addiction among young people to excessive internet use and video gaming, it's crucial to investigate how it's leading a child to detrimental on mental and physical health. Our experiment aims to examine its impact on youth mental health by collecting and analyzing relevant datasets. We gathered 780 records, dividing them into 80% training data and 20% test data. Additionally, we conducted a gender-based analysis to determine the extent of stress regarding career prospects among males and females.

After applying machine learning algorithms, we have got the expected output and the accuracy is higher than the other's algorithm. The proposed model will help to understand the impact of games on the current youth society and their situation. This study will analyze various information about the youth and help to determine their current situation.

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